

The Correlated Anomaly Observation

Five independent measurement series, one fractional deviation band

1. The observation

Measurements made in four laboratories by three independent groups, using four different detection techniques, over intervals from a single solar flare to fifteen years, report fractional deviations that fall in the same narrow band:

$$10^{-4} \leq \frac{\delta\lambda}{\lambda_0} \leq 10^{-3}$$

The series are unrelated to one another. They measure different isotopes, at different facilities, with different instruments, for different reasons. The shared quantity is the order of magnitude of the deviation, not its sign, its phase, or its period.

2. The series

Series	Facility	Interval	Fitted α	Source
³² Si seasonal	Brookhaven	1982–1986, four years	7.9×10^{-4}	Jenkins et al. 2008
³⁶ Cl seasonal	Brookhaven	1982–1986, four years	6.2×10^{-4}	Jenkins et al. 2008
²²⁶ Ra seasonal	PTB, Germany	1983–1998, fifteen years	8.3×10^{-4}	Siegert et al. 1998
⁵⁴ Mn solar flare transient	Purdue	13 December 2006, hours	$\alpha A_{\text{flare}} = -2.5 \times 10^{-3}$	Jenkins & Fischbach 2009

³² Si/ ³⁶ Cl amplitude	Brookhaven	239 data points	$7.9(3) \times 10^{-4}$	Jenkins et al. 2008
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Fitted coupling α under the model of the time-gradient field, except where noted. The ⁵⁴Mn entry is not a bare α : the source reports the product αA_{flare} , the coupling multiplied by the flare amplitude, and that product is negative. It is listed as published and is not a like-for-like value in the column. The Brookhaven series additionally reports an annual oscillation of $7.9(3) \times 10^{-4}$ with a 35(2) day phase shift, a correlation of $r = 0.52$ against $1/R^2$, and a formal probability of random correlation $p = 6 \times 10^{-18}$. The ⁵⁴Mn transient preceded electromagnetic arrival by 36–40 hours.

3. What the values share

The three seasonal fitted couplings span 6.2×10^{-4} to 8.3×10^{-4} , a ratio of 1.34 across three isotopes and two facilities. The table above lists four seasonal rows, but the fourth repeats the Brookhaven amplitude already given in the first row rather than adding a separate measurement, so there are three distinct seasonal series. The ⁵⁴Mn transient $\alpha A_{\text{flare}} = -2.5 \times 10^{-3}$ is a factor of about three larger in magnitude and is not a bare coupling. Across different isotopes, different facilities, and different physical circumstances, the couplings agree to within an order of magnitude.

No isotope-specific parameter was introduced to achieve this. The model form is fixed in advance:

$$\lambda_{\text{eff}}(t) = \lambda_0 (1 + \alpha G(t))$$

with a single coupling α per isotope and a field $G(t)$ of order 10^{-3} . The agreement across the four series is therefore a constraint on the model form, not a consequence of fitting freedom in it.

4. Reported frequencies

Power-spectrum analysis of the Brookhaven data by Sturrock et al. (2010) reports the annual frequency, a prominent peak at 11.18 yr^{-1} (a period of 32.7 days, conventionally rounded to the 33-day period) in the solar synodic rotation band, and a running-mean peak at 11.93 yr^{-1} . The 12.5 yr^{-1} frequency cited elsewhere in this body of work is reported by Sturrock, Javorsek, Fischbach, Jenkins, and Lee (arXiv:1301.3754) as a periodicity common to the BNL decay data and Super-Kamiokande solar neutrino data, described there as indicative of the solar radiative zone. The r-mode association in that literature belongs to the Rieger periodicity near 2.1 yr^{-1} , not to the 12.5 yr^{-1} line.

5. Companion series in other domains

The same band appears in two non-nuclear series, reported in the same corpus:

Series	Fractional deviation	Basis
Lunar Mascon residuals	10^{-4} to 10^{-3}	Residual after standard spherical-harmonic modelling
Apollo navigation residuals	10^{-4} to 10^{-3}	Velocity and guidance residuals on deep-space legs

6. Scope of this page

This page records an observation: that five independent measurement series report deviations of the same fractional magnitude. It does not claim a mechanism, and it does not present the field interpretation.

Every value above is quoted from published measurements or from the companion papers in this body of work, with the source named in each row. The interpretation of these deviations as a response to a scalar field is developed separately, in the time-gradient field model and its verification papers, and is not required to read this page.

7. Sources

[1] Jenkins, J.H. et al. "Evidence of Correlations between Nuclear Decay Rates and the Earth-Sun Distance." *Nuclear Physics A*, 807(1–2), 81–101, 2008.

[2] Siegert, H., Schrader, H., and Schötzig, U. "Half-Life Measurements of the Ra-226 Alpha Decay." *Applied Radiation and Isotopes*, 49(9–11), 1397–1401, 1998.

[3] Jenkins, J.H. and Fischbach, E. "Perturbation of nuclear decay rates during the solar flare of 2006 December 13." *Astroparticle Physics*, 31(6), 407–411, 2009.

[4] Sturrock, P.A., Buncher, J.B., Fischbach, E., Gruenwald, J.T., Javorsek, D. II, Jenkins, J.H., Lee, R.H., Mattes, J.J., and Newport, J.R. "Power Spectrum Analysis of BNL Decay-Rate Data." *Astroparticle Physics*, 34, 121–127,

2010.

[5] Sturrock, P.A., Javorsek, D., Fischbach, E., Jenkins, J.H., and Lee, R.H. "The Case for a Solar Influence on Certain Nuclear Decay Rates." *arXiv:1301.3754*, 2013.

Full derivations, fitted parameters, and verification: [Time-Gradient Field Model](#) and [Mathematical Verification of the Time-Gradient Field Model](#).

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AI Research Collaboration Disclosure

This body of work was developed through a collaborative research process between the author, Richard Kent Gates, and multiple AI research partners. The author provides full transparency on this process.

Author's Role (Richard Kent Gates):

- Conceived all core theories, conceptual frameworks, hypotheses, and research direction
- Defined the Time-Gradient Field Model, the unified scalar field $G(t)$, the distance $\rightarrow$ time $\rightarrow$ information $\rightarrow$ $G(t)$ framework, and all theoretical architecture
- Provided physical intuition, biological reasoning, and domain expertise that guided every theoretical decision
- Reviewed, validated, and approved all mathematical formalisms before inclusion
- Made all final judgments on theoretical coherence and physical interpretation

AI Research Partners:

- **Mimo V2.5 (opensource platform)** — Primary mathematical formalization partner. Assisted in translating conceptual physics into mathematical notation, generating numerical proof scripts, compiling LaTeX documents, and building the website infrastructure.
- **Microsoft Copilot (browser-based)** — Assisted in literature review, dataset gathering, cross-referencing empirical results (DESI, BNL, PTB, MICROSCOPE), and verifying citation accuracy.
- **Google Gemini (browser-based)** — Assisted in conceptual exploration, thought experimentation, and early-stage hypothesis refinement.
- **Space Bunny (OpenCode)** — Verification and provenance auditing. Read every paper and proof script in the repository and traced each reported figure back to the script and line that produces it. Independently recomputed the Fisher information matrices, the Cramér–Rao bounds, and the parameter errors from the

models as stated. Resolved the origin of the Fisher information values quoted in the unified and Theory-of-Everything papers, reconstructed the two-parameter Fisher matrix in the statistical verification paper from its sampling grid, and identified a mislabeled citation and a statistical criterion stated more narrowly than the reported results supported. No theoretical claim, numerical result, or interpretive judgment in this body of work originated with this tool.

Nature of AI Involvement:

The AI tools functioned as research assistants — analogous to graduate students or technical collaborators who help formalize, compute, and organize ideas that originate from the principal investigator. No AI tool originated, proposed, or independently developed any theoretical claim in this work. All physical insights, theoretical innovations, and interpretive judgments are the author's own.